Housing First or Treatment First? Ask the Kids, Not the Parents

Manuscript RED-D-26-00078

Letter to the Coordinating Editor

Summary of changes

Dear Professor Karabarbounis,

Thank you for your decision letter, and for leaving the door open to an almost new paper. That is what I am resubmitting. The model, the calibration, the validation and the experiments have all been rebuilt, and the paper has a new title. This letter summarises what changed and which comments each change answers. Four further letters respond to the Associate Editor and to each referee comment by comment, with the responses in red. Section, table, figure and footnote numbers refer to the new paper. Comments are numbered as in those letters: A1 to A20 for the Associate Editor, and R1, R2 and R3 for the referees.

What the paper now finds

The paper is now *Housing First or Treatment First? Ask the Kids, Not the Parents*, and the title is its answer. The version you rejected found that, at the same generosity, Treatment First reduced the disorder but raised homelessness, and it ranked universal Treatment First above universal Housing First only once a loss of consortium adjustment was added to welfare after the model was solved. The new paper drops the loss of consortium and asks whether moral hazard, the argument that dominates the policy debate, justifies making housing assistance conditional on sobriety.

It does not. Without any effect on children, a newborn is worse off under Treatment First by 1.09 percent of lifetime consumption, even though drug use falls and the drug test's savings buy vouchers for other households. Conditionality takes an option away from forward looking households who weigh the risk of a substance use disorder (SUD) correctly, and gives them back less through the government's budget than it takes.

The answer turns instead on the children, through three externalities that a parent counts only in part. A child shares its parent's consumption and dwelling but not the drug, so what the parent spends on drugs, and the earnings it loses while a SUD keeps it out of work, the child goes without: the budget externality. A child whose parent had a SUD while it lived at home starts adult life with lower productivity: the earnings externality. And a child lives wherever its parent lives, so a revoked voucher can put it in a shelter: the childhood homelessness externality, the one critics of Treatment First expect it to create. Within the existing rationed programme, with the drug price and the probability of drawing a voucher both clearing, Treatment First cuts drug use by 21 percent and the share of households with a SUD by 19 percent, and it lowers homelessness and the share of children in a shelter by 11 percent each. A newborn gains 2.57 percent of lifetime consumption, three quarters of it an externality its parent does not count. The parents gain too, by 0.69 percent of their own lifetime consumption, because the higher chance of drawing a voucher and the cheaper drug outweigh the option the screen removes. Along the transition, neither homelessness nor the share of children in a shelter rises in any year, and children born at the reform gain 1.45 percent.

The model agrees with the micro evidence on the households that evidence follows. Without being targeted, it comes close to the effect of making housing conditional on abstinence in the one trial that compares a sobriety condition with unconditional housing (1.40 against 1.26; Milby et al., 2005). And in the model, losing a voucher raises the three year mortality of households with serious mental illness (SMI) by 3.6 percentage points, the same order of magnitude as the 4.6 by which Housing First lowered it among homeless veterans with mental illness (Costantini, 2025). As that evidence implies, households with a SUD lose: in the comparison a trial would make, with the drug price and the voucher probability held, 74 percent of those with a SUD when the reform lands are worse off, and 67 percent are when both adjust. What reverses the verdict is what trials do not observe: the households the condition deters from ever starting, the vouchers its savings buy for others, and the children.

The channel that decides the comparison came from the reports. Referee 2 (C3) pointed to parents' concern for their children as a welfare margin the paper missed, and expected it to amplify the gain from reducing the disorder. The Associate Editor (A6) asked why the previous version discussed intergenerational transmission with no link between generations in the model. Built, the channel does more than amplify: it reverses the ranking.

What was built

A stylized model (Section 3), with two periods and calibrated (Tables 1 and 2), shows why drug use concentrates at the bottom of the income distribution when no preference varies with income (A1). Drugs are a luxury good, so without consequences the average user would earn 3.51 times the population average. The risk of dying that a SUD brings cannot be insured, and its cost grows with income; on its own it takes that figure to 0.64 at its calibrated strength. The risk of unemployment, which a household can partly insure by saving, leaves it at 3.51 even at twice its strength (Figure 1). The stylized model has saving and consequences that arrive a period later, the two things the old static appendix lacked (R1 1(e)), and footnote 10 reports the same result in a version with housing, a shelter, unemployment insurance and vouchers. **[Still open.]** Referee 1 asked for this comparison in the full model (1(d)), and the paper does not make it there. In the quantitative model the mortality multiplier of a SUD is calibrated to the income of users (Section 5.1).

A quantitative model (Section 4), life cycle and annual, in general equilibrium in the drug price and in the government's budget. Three things in it carry the paper.

- *Children.* A child lives with its parent for 25 years, shares its consumption and dwelling, and enters the parent's flow utility at the weight ω_c. A SUD the parent has while the child lives at home lowers the child's productivity for life, and the parent is charged for that loss, at the same weight, as a difference against a child whose parent never had the disorder (Sections 4.2, 4.8 and 4.10).
- *Preferences that are the same in every state.* The parameters of utility do not vary with the disorder, with SMI or with housing (Sections 4.2 and 4.7), so reducing the disorder no longer raises the utility of consumption and housing mechanically, which Referee 2 (C5) and Referee 3 (M2(a)) objected to. The weights are invariant to the disorder, as the Associate Editor suggested (A8), and so is b. The disorder works through what it costs: mortality, employment, the onset of SMI and the child's productivity. A child inherits a worse productivity draw, not a taste for drugs.

- *Recovery that depends on housing.* Referee 2 (C1), the Associate Editor (A13) and Referee 1 (2(d)) asked for discipline on the response of drug use to housing access. In the model a household with a SUD is less likely to recover in a shelter, by a factor calibrated to Milby et al.'s (2005) finding that housing raises the chance of recovering by 47.3 percent (the model gives a ratio of 1.45 against 1.47). It is a transition probability rather than a taste, and it is what the trial measures. The Associate Editor's second elasticity, of homelessness to the disorder, runs through employment: a SUD makes a job harder to find, by a factor calibrated to the SUD share of the homeless (Section 5.1, Block 2, and below).

What was cut, and what was added

The Associate Editor asked whether a list of ingredients earned their place. I cut or simplified each of them but one, the link between generations, which the paper now models and which carries the result. Earlier versions solved the model with endogenous labour supply, rents, homeownership and house prices, and these margins had a negligible effect on the results (footnote 19).

Ingredient	Now	Comments
Homeownership, mortgages and house prices	Cut. Households rent. The grid is the shelter and three rental sizes; the shelter and the smallest rental are calibrated, and the two larger sizes are set at four and eight times the smallest, with no moment and no sensitivity (Table A1).	A4; R2 D1
Endogenous rents	Cut. Equilibrium is now where the Associate Editor asked for it, in the government's budget and the drug price, and Table 5 switches each on and off.	A5
Labour supply	Cut. Labour is supplied inelastically when employed.	A3; R3 m1
Unemployment insurance	One year of benefits for the share of job losers who are insured, nothing after (Section 4.3).	A2
Type dependent transfers and bequests	One endowment at entry, the same for every entrant, from the estates of the dead; the government collects the rest.	A7
Fiscal costs of homelessness and of a SUD	Cut. The shelter is free, and the budget counts what the government pays in transfers, drug tests and treatment (Section 4.11).	A15; R2 C6
Light and heavy use	Cut. One chance that a year of use leads to a SUD (footnote 20).	
Loss of consortium	Cut (below).	A9; R1 3(a) to 3(c)
Universal vouchers	Cut. Every experiment keeps the programme rationed (below).	A18; R2 C6; R3 m3

Against that, the model gained the following, each at a report's request.

1. **SMI** (R1 1(a); Section 4.7): a persistent state whose onset a SUD makes twice as likely, and which lowers the chance of finding a job, raises mortality and slows recovery from a SUD. Without being targeted, households with SMI are 2.64 times as likely as others to use drugs, against 2.44 in the data. Referee 1 expected it to weaken Treatment First; the screen does fall hardest on households with SMI, and Treatment First still raises welfare. **[Still open.]**

The paper does not report a run without SMI, so it does not measure how much SMI weakens Treatment First.

2. **Treatment** (R1 1(b); A10; Section 4.6): a choice, at a utility cost calibrated to a take up of 24.5 percent, that doubles the chance of recovering and that the government pays for. Under Treatment First a household with a SUD passes the screen only in treatment and without using, and the share in treatment rises from 24.6 to 29.0 percent. If a household with a SUD cannot pass the screen at all, Treatment First still lowers homelessness by 11 percent and a newborn gains at least as much (footnote 38).
3. **An equilibrium drug price** (R1 1(c); R3 M2(d); A5; Section 4.12), clearing a supply curve with an elasticity of 1.46, Mace et al.'s (2020) estimate for cannabis; the long run elasticity of an illicit supply is likely higher, which would shrink the price effect (footnote 31). It works against Treatment First: the price falls by 5 percent and draws users back, which takes the newborn's gain from 6.45 to 2.57 percent.
4. **Imperfect and costly enforcement** (R1 4(a) and 4(c); A11; Sections 4.9 and 6). The screen catches a user with probability 20 percent a year, set so that it deters about as much as the US military's random testing did: 62 percent of the users facing it stop, against a fall of 61 percent in past year marijuana use (Mehay and Pacula, 1999). That 61 percent is the raw comparison among service members who stayed; counting those who left, the fall was 11 to 45 percent, and at half the detection rate 41 percent of users stop (footnote 33). A test costs 35 dollars, the upper end of the estimates, and a household that is caught is excluded for three years on average. At half the detection rate, Treatment First still cuts drug use by 14 percent and homelessness by 6 percent, and a newborn gains 1.65 percent (footnote 35).
5. **Transitions** (R1 3(d); Sections 6.4 and 6.5). The transition starts from the whole Housing First population alive at the reform and clears the drug price and the voucher budget every period (Figure 3). Drug use falls on impact, 81 percent of its long run fall in the first year, and homelessness and the share of children in a shelter never rise; 11 percent of the adults alive at the reform are worse off. When a household with a SUD cannot stop using (Table 7), the design matters: a screen that catches every user from the first day raises homelessness by up to 14 percent, while at the calibrated detection rate, exempting households in treatment for the first five periods holds the rise to 1.6 percent and keeps the share of children in a shelter from rising at all. Children born at the reform gain in every case.
6. **The productivity cost of a SUD** (A20). **[Still open.]** There is no direct wage penalty on the user: the disorder lowers a user's earnings by keeping it out of work longer (Section 4.8). The productivity cost the model carries is on the child, set to 0.24, Nilsson's (2017) natural experiment estimate and the closest causal one, although its shock, alcohol in the womb, does not map one to one into a parent's disorder after birth (Section 2). Even at zero, Treatment First raises welfare (footnote 5).

The loss of consortium adjustment is gone

Referee 1 (3(a) to 3(c)) and the Associate Editor (A9) objected to it: an adjustment applied after the model was solved, with no moment behind it, which changed the ranking, and whose conservative value is arguably zero. I agree, and it is deleted. Welfare is now a newborn's own expected lifetime utility, childhood included (Section 6.1), and nothing is added to it after the model is solved.

What decides the ranking now lives inside the household's problem. A parent counts the share ω_c of what its choices cost its child and acts on it; the rest is the externality. The weight is calibrated to behaviour: in the 2024 National Survey on Drug Use and Health (NSDUH), unmarried adults living with one other adult and no child are 17.5 percent more likely to have used illicit drugs in the past year than comparable lone parents, and ω_c is the weight at which removing a parent's concern for its child raises drug use in the model by as much (Section 5.1, Block 5). At Nishiyama's (2000) weight of 0.69, the upper end of the literature, a newborn gains more, 3.18 percent, though less of it is an externality (footnote 28). Section 6.3 makes the externalities appear numerically, as Referee 3 asked in M2(c): it splits the newborn's gain into the childhood externality, 1.50 percent of lifetime consumption, the earnings externality, 0.46, the newborn's own adult life under the policy, 0.43, and the part its parent counts, 0.19 (Table 6).

On tone (Referee 1, 3(c)), the paper says who loses as plainly as who gains, and its conclusion is conditional: yes if the externalities on children are counted, no if the case rests on moral hazard (Section 7). In the long run 74 percent of households are better off and 3 percent worse off, and 24 percent of those with a SUD are better off (Section 6.2).

[Still open.] The Associate Editor suggested reporting welfare and mortality separately. The paper does not report mortality as an outcome of the experiments. Mortality enters welfare through the value households place on their own lives, calibrated to the value of a statistical life, and the paper checks the model's mortality response against Costantini (below).

Errors in the previous version

[Conceded.] Rebuilding the calibration from the data turned up three errors in the version you rejected.

1. **The share of the homeless with a SUD**, the 37 percent the Associate Editor questioned (A16), was the midpoint of the range of substance abuse rates, about 25 to nearly 50 percent, in Kuhn and Culhane (1998), not a rate of drug use disorder. The paper now targets 21.7 percent, the drug use disorder share of the homeless in Gutwinski et al. (2021), and the model gives 20.2. Homelessness means literal homelessness, a shelter or the street (Section 4.5).
2. **The unemployment rate given a SUD was too high.** The previous version targeted 26.9 percent, from 2021 and 2022. For adults aged 26 and over, the ages the model covers, the 2024 NSDUH gives 16.2 percent. It is no longer a target, and the model gives 14.5 percent (Table 4).
3. **Drug use, the income of users and drug spending came from different definitions of a user.** A user is now an adult aged 26 or over whose use amounts to a drug use disorder of any severity, 8.90 percent in the 2024 NSDUH, and a household with a SUD is one whose disorder is moderate or severe. The income of users and the share of users with a SUD are measured on that population, and drug spending is taken over its income (Section 5.1, Block 1).

Calibration and validation

Referee 1 could not assess the quantification because the calibration table never reported the model's moments (2(a), 2(b)), and the Associate Editor agreed (A12). Of the model's 59 parameters, the 26 that carry the story are in Table 3, in six blocks, and each entry says what the parameter controls, how it bears on the two policies and what happens as it rises, before its value, moment and source (A17); the hazards of death and unemployment are set out the same way, each with its source or moment (R3 M1(b)). The 33 standard ones are in Appendix A (Table A1). Both tables report the data, the model and the value for every target. The replication package documents the dataset, year, universe and computation behind every estimate, and carries the data (A17; R1 2(e)). Homelessness is targeted as a band of 0.40 to 1.00 percent of households, from the Department of Housing and Urban Development's single night count to the share homeless at some point in a year (Burt et al., 2001), and the model gives 0.62 percent (A16). **[Still open.]** Baseline recovery is still set from the ten year remission rate in Vaillant (1995), a study of alcoholism, and the paper does not argue that it carries over to a drug use disorder among the homeless, which the Associate Editor questioned (A17).

Most targets are close. The largest gaps are the income of users, 62 percent of the average in the model against 74 in the data, the share of renters paying more than half their income, 25.8 against 19.2 percent, and the unemployment rate, 5.6 against 4.9 percent. **[Conceded.]** The income of users is the gap Referee 1 raised in 2(c)(i), where the previous model put it at 41 percent. Referee 1 was right that the model made users too poor. Their income still falls short of its target, but by less. The calibration targets only the average income of users, but the model generates a distribution of drug use by income in line with the data, which a new figure sets against the 2024 NSDUH (Figure 2; R3 M1(c), and in part M2(b)): use is highest at the bottom and lowest at the top, as in the data, while in the model it is higher than the data at the bottom and flatter and lower in the middle, and both lower the income of users. Section 5.2 says why the income and age of users matter for the experiments: both policies reach a child through its parent, and the vouchers are means tested.

[Still open.] The model still misses on the homeless (R1 2(b), 2(c)). They are poorer and younger than in the data: 21.4 percent are employed against 46, their income is 6.5 percent of the average against 19.1, and 11.5 percent are 55 or older against 25.9 (Table 4, Block 1). Users are younger too, a mean age of 34.6 against 39.6. Section 5.2 states each gap beside the data and says why that margin matters for the experiments.

Four of the policy responses the reports asked the model to match (R1 2(d); R2 C1; A13) are now in the paper. Two are calibrated: the response of recovery to housing, to Milby et al. (Table 3), and the response of drug use to a drug test, to the military's testing (Section 6). Two are not, Milby et al.'s effect of making housing conditional on abstinence and Costantini's mortality effect, and the new validation section (Section 5.2, Table 4) shows the model comes close to both (above). **[Still open.]** The paper does not compare the response of homelessness to voucher access, or of employment to vouchers, with a trial estimate (R1 2(d)). Section 5.2 also compares the homeless, users and voucher holders in the model with the data (R3 M1(a)), though not drug use among voucher holders. The price elasticities are no longer targets (R1 2(e); R3 M2(d)): participation responds more than in the evidence, -1.32 against -0.3 to -0.9, which works against Treatment First, since the price it lowers draws users back. Transmission across generations is at the low end of the evidence for a SUD, an odds ratio of 1.46 against 1.44 to 2.51, and above it for homelessness, 6.23 against 1.8 to 2.4, so the model loads the effect on children towards the margin Housing First

acts on.

Costantini (2025)

The Associate Editor (A14) read Costantini as finding that Housing First lowers the three year mortality of homeless veterans by 4.6 percentage points while Treatment First has no lasting effect on health, in contrast with the model. Validation Block 7 now compares the model with Costantini's Housing First estimate. With the drug price and the voucher probability held, losing a voucher raises the three year mortality of the households with SMI whom Treatment First's revocation puts on the street by about 3.6 percentage points, against Costantini's 4.6, the same order of magnitude, without being targeted.

[Still open.] The paper compares the model with this arm only. Costantini's Treatment First arm is the Department of Veterans Affairs' own programmes, which the model does not represent, since its Treatment First is a voucher whose retention depends on sobriety (Section 4.9), and I do not claim a counterpart to it. The paper also does not cite Cohen (2024), Brounstein and Wieselthier (2026) or Humphries et al. (2026); of the structural work, it cites Abramson (2026), on eviction policy.

So the model agrees with him that losing housing raises mortality for the households who lose it. What that design does not observe is everyone else: in the long run Treatment First lowers homelessness by 11 percent, so fewer households are on the street at all.

Where I push back, and what the paper does not do

Costantini is the first place, above. The others:

1. **[Still open.] Labour supply** (R3 m1). It is gone (above), so the paper has no labour supply elasticity. Referee 3 expected one to matter through the cost of balancing the budget with labour taxes; taxes stay at their benchmark values in every experiment and the voucher probability balances the budget, so no labour tax moves.
2. **[Still open.] The age limit on starting to use** (R1 S1). It stays, at age 80, and stops only households without a SUD from starting. Without it, households close to the model's last age, who know how little life remains, use too much (footnote 21). In the model 2 percent of users are 65 or over, against 10 percent in the data (Table 4).
3. **[Still open.] Generosity.** Referee 2 asked about optimal voucher design (C6), Referee 3 for a continuous experiment in the voucher probability (m3) and for how optimal policy changes when a channel is shut down (M2(c)), and the Associate Editor found the rationed and universal comparisons the most relevant (A18). The paper now asks one question, conditionality, within the existing rationed programme. It does not study how generous the programme should be, and it has no universal programme.
4. **[Still open.] False positives** (R1 4(b)). The model has none, because a positive screen is confirmed in a laboratory before it has consequences (Section 6), and there is no sensitivity to them. Stigma and administrative burden (4(c)) are not modelled beyond the cost of a test, set at the upper end of the estimates.
5. **[Still open.] Fiscal costs** (A15; R2 C6). The model no longer has a fiscal cost of homelessness

or of a SUD, so the paper makes no comparison with Abramson and Van Nieuwerburgh or with the Council of Economic Advisers, and has no robustness run on them.

6. **[Still open.] The disamenity of homelessness** (R2 C2). It is not in the benchmark, and Corbae, Glover and Nattinger (2024) is not cited. With one at Loubière et al.'s (2020) contingent valuation, about 0.2 percent of average earnings a year, no long run gain in Table 5 moves by more than 0.14 percentage points (footnote 4).
7. **[Still open.] Identification and sensitivity.** The paper reports neither the numerical identification Referee 1 asked for (2(f)) nor a systematic sensitivity analysis (4(d)). It reports five robustness runs: half the detection rate, the parent's weight at the upper end of the literature, no productivity penalty for the child, households with a SUD unable to pass the screen, and a disamenity of homelessness. Treatment First raises a newborn's welfare in each.

Presentation

The body carries one comparison, Treatment First against Housing First in the rationed programme, taken apart step by step in Table 5, decomposed in Table 6 and followed along its transition in Figure 3 and Table 7. No experiments sit in an appendix, which holds only the standard parameters, and the robustness runs are reported in footnotes (A18). The word "addicted" is gone (A19). Table 5 reports every outcome in levels under each policy, and the text gives the changes in percent (R1 S2). The typographical errors Referee 1 listed (S3 to S10) were in passages and references that have been rewritten or removed. **[Still open.]** There are no plots of the drug use policy function (R2 C4) and no sensitivity graphs (R3 m2); the paper shows who uses, and why, through Figures 1 and 2.

Thank you again for the second look.

Sincerely,

Shahar Rotberg